

Smart Real-Time Age and Emotion-Based Advertising in Malls Using Custom-Built Models

Project ID: 25-26J-091

Project Proposal Report

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Bachelor of Science Special Hons Degree in Data Science

Sri Lanka Institute of Information Technology

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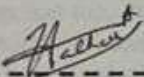
August 25

DECLARATION

I declare that this is my own work & this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of my knowledge & belief it does not contain any previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research section is focused on developing an interactive game-based module for real-time age detection to enhance targeted advertising in shopping malls. Traditional advertisement displays in retail spaces follow pre-scheduled time schedules with fixed content, thus low audience interaction and low personalization. To address this, the proposed system integrates concepts of computer vision, machine learning, and gamification into a single platform which not only determines the demographic profile of the viewers but also actively engages them through interactive elements.

The module operates by inviting consumers to participate in a quick on-screen game on digital signage. While the game is serving to attract the attention of consumers and maximize dwell time, the system is also simultaneously using deep learning–facilitated facial recognition to estimate the age of the player. These age impressions are fed into a real-time advertisement suggestion engine, which selects and displays content that is most appealing to the inferred demographic category. This approach increases the effectiveness of age detection—since better frontal facial captures are obtained through play—and performance of ad delivery, since engagement yields better-quality data.

The end-to-end system has three interconnected layers: (1) emotion and demographic recognition by privacy-protected facial embeddings, (2) interactive engagement interface to engage and attract users, and (3) dynamic ad content adjusting recommendation engine by the majority audience profile. The system has GDPR-compliant data practices integrated into the design, where it does not retain raw images and processes all personal features in real time to preserve privacy.

This unified platform bridges the gap between passive audience identification and active user engagement, creating an adaptive advertisement platform that is both effective and privacy conscious.

Keywords: Interactive advertising, gamification, age detection, targeted marketing, facial analysis, real-time recommendation, privacy-preserving AI, deep learning.

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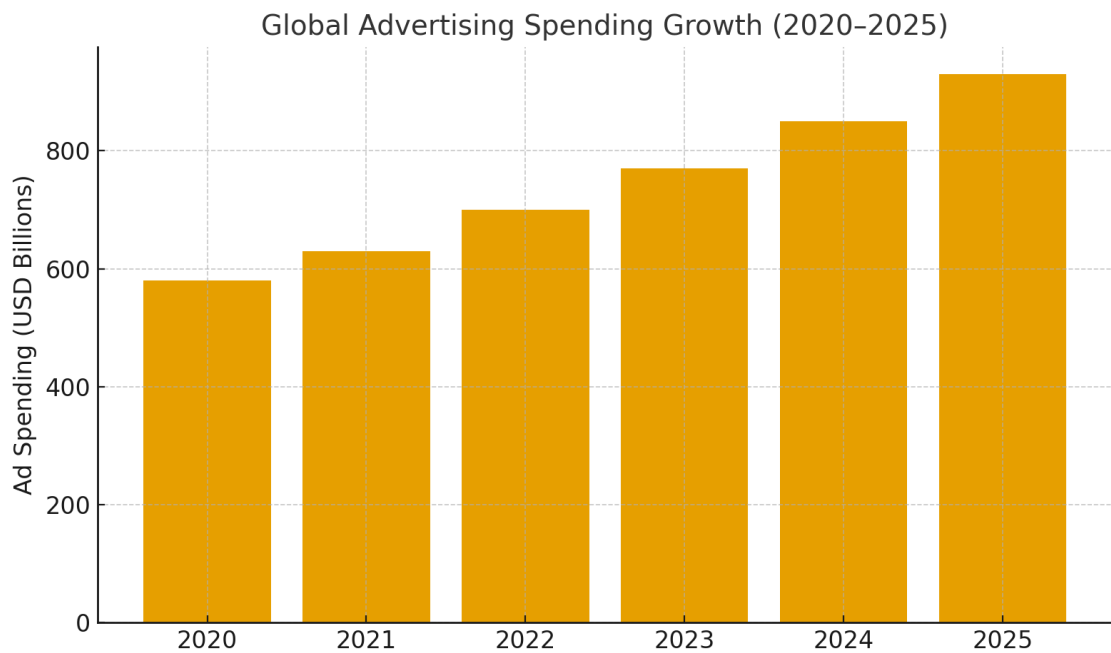
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LIST OF ABBREVIATION

Abbreviation	Full Form	Description
AI	Artificial Intelligence	Field of computer science concerned with creating systems that can perform tasks requiring human-like intelligence.
AR	Augmented Reality	Technology that overlays digital information (images, objects, effects) onto the real-world environment.
CNN	Convolutional Neural Network	A deep learning architecture commonly used for image and facial recognition tasks.
CRM	Customer Relationship Management	A system for managing interactions with current and potential customers.
DOOH	Digital-Out-Of-Home	A form of outdoor advertising delivered through digital screens (e.g., in malls, airports).
GDPR	General Data Protection Regulation	European Union regulation for data protection and privacy.
GPU	Graphics Processing Unit	A processor specialized for parallel computing, widely used in AI model training and inference.
HCI	Human-Computer Interaction	Study and design of interfaces between people and computers.
HOG	Histogram of Oriented Gradients	Feature descriptor used in computer vision for object detection.
IP	Internet Protocol	Communication protocol for devices over a network, here referring to IP cameras.
KPI	Key Performance Indicator	Measurable values used to evaluate system or project success.
LBP	Local Binary Patterns	Texture descriptor used in facial recognition and image analysis.
PII	Personally Identifiable Information	Any data that can identify an individual (e.g., name, face image).
YOLO	You Only Look Once	A family of fast real-time object detection models, here adapted for facial detection.

1. INTRODUCTION

With the rapidly evolving retail landscape, shopping malls remain center-stage locations for consumer activity, offering an array of products and experiences in one venue. As competition rises among brands, the ability to deliver customized, real-time, and engaging commercials has become a key differentiator in influencing customer decisions. With the growing popularity of technology among young adults and adults alike, shopping malls now find it necessary to adopt innovative, technology-driven methods of advertising to be noticed and stay relevant. Most traditional advertisement display platforms in malls rely on static algorithms, which sequentially disseminate pre-defined content at set intervals without making any allowances for demographics, interests, or the context of the viewers now. This "one-size-fits-all" approach consistently results in useless ad placements, decreased customer engagement, and lost opportunities for advertisers to target potential buyers effectively.



The proposed system seeks to address these deficiencies by integrating leading-edge computer vision and machine learning techniques to transform mall advertising into an intelligent, real-time recommendation platform. The primary function relies on sensing and understanding multiple parameters of the waiting audience in front of the advertisement screen—i.e., age, gender, emotion, behavior, and carried objects—and using this knowledge to select the most suitable ad dynamically from a pre-filtered database. By personalizing ad copy to match the audience in front of it at any given time, the system seeks to dramatically boost relevance of ads, drive greater audience engagement, and achieve greater sales impact. The system is built to reach both unique and mass audiences, so brands can reach on an individual basis while appealing to the larger group too, thus driving greater overall marketing effectiveness.

One essential component of this research is the "Interactive Game-Based Age and Gender Detection for Targeted Advertising" A portion of this aspect is the encouragement of audience engagement through the integration of a simple but interactive game as a part of the advertisement screens in shopping malls. The game includes direct interaction with the screen, the entertainment of which is a friendly and unobtrusive method of collecting demographic information such as age and gender. Following play, the system delivers player-specific advertising to the individual player, followed by content targeted to the broad mass audience surrounding them. Balancing personal relevance with broad audience appeal are accomplished through this two-step targeting. With the integration of interactive entertainment and demographic ad targeting, this section will transform passive advertising screens into active, dynamic audience-reactive systems that enhance the shopping experience for consumers as well as measure campaign effectiveness for advertisers.

With the integration of engagement-driven interaction, real-time audience analysis, and privacy-conscious design, this research is expected to redefine the process of showing

advertisements by shopping malls. The suggested system bridges the existing gap between physical retail spaces and intelligent digital advertisement, enabling advertisers to have better targeting accuracy and customers to enjoy better, more relevant, and engaging advertising experiences.

1.1. Background

1.1.1. Physical Retail Challenges of Personalization

Increasing sophistication in digital marketing has made personalization a top driver of ad performance. In the physical shopping environments like malls, though, personalization has lagged online media due to the challenges in gathering, analyzing, and responding to customer data in real time. Traditional mall ad systems tend to display the same static messages in rotation, regardless of the individuals in front of the screen. While it's simple to implement this method, it is not quite flexible and results in substandard audience engagement as well as lower rates of conversion.

1.1.2. Technological Advances Enabling Intelligent Advertising

The recent developments in computer vision and machine learning have made it possible to have smarter and context-sensitive ad systems. They help examine audience characteristics in real time so that ads are targeted to a more specific demographic and emotional makeup of the audience.

1.2.Literature Survey

1.2.1. Traditional vs. Modern Approaches in Audience Estimation

Traditional manually designed feature-based works in age estimation relied on Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG). The traditional approaches were though vulnerable to lighting variations, pose changes, and occlusions, which limited their applications in real-world scenarios.

With the development of deep learning, namely convolutional neural networks (CNNs), robustness and accuracy have significantly improved and become more feasible for real-time deployment in public spaces. Similarly, gender classification models have evolved from simple binary classifiers to sophisticated deep neural network architecture with capabilities to achieve high accuracy even in poor conditions.

1.2.2. Machine Learning Techniques in Demographic and Emotion Detection

Apart from basic demographic identification like age and gender, researchers have looked at emotion recognition as one important variable for targeted marketing. Analysis of facial expressions can say a lot about a person's state of mind in the present, which can influence consumption behavior. Dynamically modifying ad content based on emotion detection has been executed in certain research. Such systems overlook other important context clues like audience behavioral patterns or objects that are being held, which may further boost ad targeting.

1.2.3. Gamification and Interactive Engagement in Retail Environments

Gamification has globally been recognized as an effective tool to attract attention and encourage engagement in retail environments. Previous implementations have utilized touchscreens, interactive kiosks, and augmented reality (AR)-based games for increasing customer linger time and improving user experience. Being effective or not, incorporating gamification with real-time demographic analysis for demographics-based advertising remains an untapped territory. There are current systems that are interested in entertainment without linking interaction to personalized content, or passive demographic identification without active engagement of audiences.

1.2.4. Privacy-Preserving Techniques of Public Facial Analysis

As facial analysis technologies are increasingly used in public places, privacy concerns have come into prominence. There is growing interest in techniques that do not store raw facial images to ensure individual privacy. New approaches leverage facial embeddings—small vector summaries of facial features—enabling identification and demographic inference in a manner that does not store pictures with identifiers. Such privacy-conscious method aligns with global data protection regulation such as the General Data Protection Regulation (GDPR), which demands stringent controls of processing personal data.

1.2.5. Cohesive Systems for Real-Time, Audience-Aware Advertising

While demographic identification, emotion detection, and gamification have each been studied in immense detail independently, there is certainly a need for end-to-end systems that combine these into a single, real-time, audience-aware platform for advertising. Additionally, current solutions seldom address the challenge of engaging the audience

proactively in a way that enriches data quality as well as benefits the user experience. This research tries to make up for this lack by developing an integrated system that is well-suited to shopping mall environments, combining real-time audience measurement, interactive engagement, and privacy-aware design guidelines.

Table 1- Literature Review for the whole system

Feature / Capability	Zhang et al. (2017) – RoR CNN Age/Gender	Smith & Chen (2018) – Transfer Learning	Islam & Baek (2021) – Smart Store CRM	Terlutter & Capella (2013) – Gamified Ads	Morales et al. (2019) – SensitiveNets	Aslan & Çelebi (2022) – Real-Time Ads	Proposed System – Interactive Game-Based Age & Gender Detection
Real-time Age Detection	✓	✓	✓	✗	✗	✓	✓
Gender Detection	✓	✓	✓	✗	✗	✓	✓
Game-based Engagement	✗	✗	✗	✓	✗	✗	✓
Advertising/Marketing Focus	✗	✗	✓	✓	✗	✓	✓
Personalization / Recommendation	✗	✗	✓	✓	✗	✓	✓
Privacy-Preserving Techniques	✗	✗	✗	✗	✓	✗	✓
Retail / Mall Application	✗	✗	✓	✓	✗	✓	✓
Serendipity / Novelty in Ads	✗	✗	✗	✗	✗	✗	✓
Integration of Multiple Modalities (age, gender, mood)	✗	✗	✗	✗	✗	✗	✓

1.3. Research Gap

Table 2- System Comparison Between Existing Systems and Proposed Application Component

Feature / Objective	Existing Systems (Research)	Proposed Application Component
User Engagement	Facial analysis systems are typically passive; viewers are not motivated to look directly at the display.	Use an on-screen interactive game to attract attention and improve dwell time, resulting in better data capture.
Age Detection Accuracy	Conventional systems suffer from poor captures due to side views, pose variations, or lack of direct gaze.	Gameplay naturally encourages frontal facial orientation, improving deep-learning age estimation accuracy.
Link to Personalization	Gamification in retail is usually for entertainment only, unrelated to ad targeting.	Integrate gamified interaction directly with ad recommendation to deliver content relevant to detected age groups.
Privacy Handling	Some existing models store raw facial images or do not comply with privacy regulations.	Use facial embeddings instead of raw images for processing, ensuring GDPR-aligned, privacy-preserving design.
Real-Time Adaptation	Many systems cannot update ads dynamically; targeting decisions are precomputed or slow.	Fully real-time age-aware recommendation engine that adapts ad content as demographics change.
User Experience	Passive surveillance approaches can feel intrusive or unwelcoming.	Make advertising interactive and playful, creating a positive user experience rather than one-sided monitoring.

While age estimation, gamification, and privacy-friendly techniques were each explored individually, they rarely are combined in a single system for customized advertising in physical stores. Existing solutions belong to either passive (age estimation with accurate audience counts without engaging audience participation) or interactive types (games or augmented reality experiences without demographic analysis connection). There is no existing research that fully combines real-time age estimation, gamefic engagement, and privacy-friendly design to boost data quality, user engagement, and ad customization simultaneously.

This research fills the gap with an audience-conscious advertising component where the engaging game serves a dual function: to entertain the users and, in the background, high-quality demographic data collection for the recommendation engine without infringing privacy.

This table is a recap of the existing system and research gaps and how your solution aims to fill them with goals, including gamified engagement, real-time age estimation, and privacy-guarding design.

1.4.Research Problem

Within modern shopping malls, electronic billboard advertising screens are a primary brand promotional medium, yet the overwhelming majority operate to fixed schedules or programming of repetitive loops of one static set of ads to every viewer. This one-size-fits-all approach does not consider the demographic make-up of the real-time viewership, resulting in less viewership engagement, diminished content relevance, and ultimately reduced conversions. With retail spaces becoming increasingly competitive, there is great need for ad systems that will be able to adapt dynamically to the people who come before the display.

Recent developments in computer vision and machine learning have allowed for the use of facial analysis to estimate demographics such as age and gender. Most existing systems, however, are passive and rely on the observers briefly looking at the display screen. The infrequent or skewed views do not work as well and tend to yield partial or inaccurate demographic data, decreasing targeted-ad success. On top of this, there are many implementations that are bringing serious privacy concerns by holding raw facial images or personal data without applying methods that would adhere to legislation such

as the General Data Protection Regulation (GDPR). As a result, while technology for age detection has been developed, its uptake by advertisement systems has been slow, sporadic, and invasive in most instances.

Gamification has emerged as an effective means of grabbing consumer attention and elevating the time spent interacting in the retail environment. Interactive kiosks, touchscreens, and AR-based games have been deployed to entertain customers and create memorable brand experiences. These gamified elements are rarely connected to ad personalization, however. Existing systems are either focused on precise demographic detection without engaging the users actively, or on user engagement without leveraging the interaction to target content of ads. This compartmentalization of functionality results in opportunities lost demographic information continues to go untapped, and interactive experiences cannot be parlayed into targeted marketing advantages.

This study addresses these constraints by developing an interactive game-based age detection module that combines engagement, personalization, and privacy in a single system. By embedding the age estimation in an on-screen game, the solution organically encourages consumers to look straight at the screen and actively interact with it, resulting in clearer, frontal facial captures and improved detection accuracy. The age information collected during gameplay is processed using privacy-preserving facial embeddings rather than raw images, which complies with global data protection regulations. It is then fed into a real-time recommendation engine, which dynamically selects adverts most relevant to the detected demographic group.

By doing this, the work demonstrates how gamification can deliver a dual advantage: introducing entertainment value into advertising and improving the quality of demographic information available for personalization. Unlike existing systems that

address engagement and personalization separately, this combined solution offers an intelligent, adaptive advertising platform calibrated to shopping mall environments—one that delivers targeted content, guarantees privacy, and transforms advertising into an interactive experience from passive display.

2. OBJECTIVES

2.1. Main Objectives

The primary objective of this research is to design and develop an interactive game-based module that is capable of detecting shoppers' age in real-time for enabling dynamic, targeted advertising for shopping malls. Unlike conventional advertising systems, which offer pre-programmed content regardless of the demographic profile of the audience, the system to be proposed will integrate user interaction and demographic analysis. With the incorporation of gamification, the module attempts to take interest, encourage active participation, and capture frontal facial images that boost age detection accuracy. Apart from this, this system will focus on privacy-conscious design by using facial embeddings rather than raw facial images so that it can maintain compliance with global data protection laws such as GDPR. The ultimate goal is to create an advertising platform which not only dynamically personalizes content but also enhances user experience and public confidence.

2.1. Specific Objectives

- **To design and develop a two-way interactive game face to involve the audience and facilitate proper face orientation.**

Traditional facial analysis systems struggle to obtain high-quality pictures because observers are not looking at the screen. By integrating age detection into a simple, fun, and quick game, this study aims to make people automatically face their faces toward the camera and, hence, enable more accurate data to be captured. The game will be made easy, innocuous, and quick to ensure that it is suitable for busy shopping malls.

- **To create an effective age-detection model through deep learning optimized for real-time use in public spaces.**

The system will utilize convolutional neural networks (CNNs) or other cutting-edge architectures capable of detecting age accurately even in situations of fluctuating light, face pose, and partial occlusions. The model will be trained and optimized so it will execute well on edge devices or networked platforms without sacrificing speed, thereby becoming viable for deployment within shopping malls.

- **To incorporate privacy-preserving techniques that process demographic information without holding personally identifiable details.**

Since face information is confidential, the system will use face embeddings rather than raw image storage. This reduces the privacy risks without compromising on the ability to perform accurate age estimation. The module will be GDPR and other data protection legislation compliant so that no confidential user information is stored or exchanged except for required real-time processing.

- **To integrate the age detection module with a dynamic ad recommendation engine for real-time content delivery.**

The age detection results will be fed into a recommendation system that selects and delivers ads suitable to the identified demographic segment. The integration will render delivered content more relevant, improving engagement and conversion rates over the traditional loop-based advertising practices.

- **To calculate the system performance in terms of age detection accuracy, user experience, and advertising efficiency.**

The resultant system will be analyzed for whether gamification increases substantially the accuracy of age detection and whether dynamically targeted advertisements result in enhanced response from the audience. Such metrics as precision of detection, dwell time, and ad relevance will be monitored and compared to existing static or passive systems to verify the benefit of the proposed strategy.

3. SYSTEM METHODOLOGY

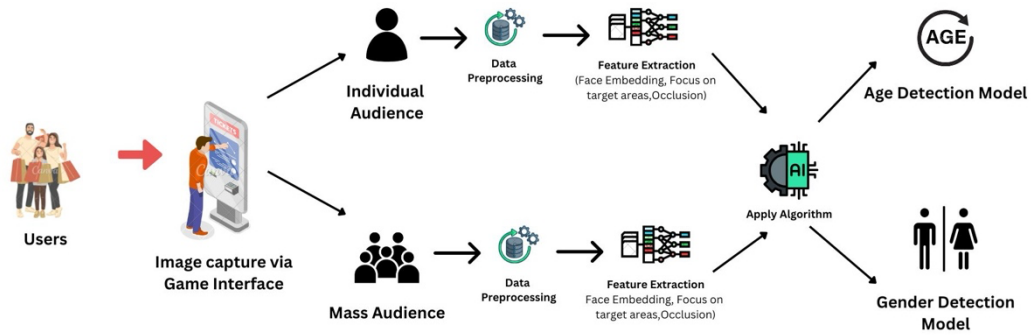


Figure 2-Overview of the Interactive Game Based Custom Built Age and Gender Detection System

The strategy for this research step is to integrate human–computer interaction principles with machine learning technology to enable a privacy-friendly, interactive system for real-time age estimation and context-aware advertising in shopping malls. Unlike conventional passive static facial recognition methods that only take photos, this technique applies an interactive game-based platform to encourage active engagement by users, offering superior quality facial information and more effective detection. The approach establishes step-by-step procedure from conceptual design to deployment, with iterative testing, data-driven optimization, and ethics. It incorporates the use of computer vision models optimized for edge deployment, the application of privacy-focused mechanisms such as facial embeddings, and development of a dynamic ad recommendation pipeline. Through systematic phases of requirements analysis, system design, model construction, validation, and field testing, this methodology assures the system to be robust, scalable, and robust under divergent environmental conditions while

fulfilling the objectives of accuracy, response time, and user acceptability.

3.1. Application of Key Pillars within the Specialized Knowledge Domain

This study makes use of computer vision, machine learning, human-computer interaction (HCI), and privacy-preserving AI concepts.

- Computer vision provides the foundation for face detection and capture in real-time. Techniques like face detection, alignment, and embedding generation allow the system to process faces without storing raw images.
- Machine learning, particularly deep learning architectures like convolutional neural networks (CNNs), enables accurate age estimation even in challenging environmental conditions like varying illumination or partial occlusion.
- HCI and gamification ensure that the users are engaged long enough to provide quality facial data. Interactive nature also causes the advertising experience to shift from passive viewing to active, engaging entertainment.
- GDPR and other regulations-compliant handling of data is supervised by privacy-safeguarding AI through the use of vectorized representations (facial embeddings) instead of storing raw images in a direct manner.

By synthesizing these columns, the framework bridges technology innovation and user-centric design and ethics-oriented approach.

3.2. Application of Technologies in the Field of Application

The proposed module will utilize current frameworks and hardware to provide real-time, high-precision performance:

- **Age detection and computer vision:** PyTorch or TensorFlow will be used for training and deploying CNN-age detection models. Pretrained face detection (e.g., YOLO-face, RetinaFace, or MTCNN) will allow for robust and efficient face localization.

- **Gamification interface:** A minimal front-end platform (with web-based HTML5 and WebGL or Unity or React) will create an engaging mini-game running on kiosk systems or digital signage.
- **Privacy and data processing:** Facial embeddings will be computed with models like FaceNet or ArcFace so that raw images never have to be saved. Data will be processed locally at the edge (e.g., with NVIDIA Jetson hardware or similar embedded GPUs) to minimize risks in transmission.
- **Real-time recommendation engine:** A back-end service using Flask or FastAPI can compute demographic data to select and serve ads in real time.

These technologies are chosen based on their trade-off between computational efficacy, scalability, and relevance to real-time retail application.

3.3.How to Carry Out the Project

The project will adopt an iterative development life cycle:

- Requirement analysis and design – Define technical requirements for the game interface, age detection model, and recommendation logic.
- Prototype development – Create the first version of the game and integrate a pre-trained age detection model to test feasibility.
- Model training and optimization – Optimize the age detection model on a set of images to improve performance in mall settings.
- Privacy integration – Embedding-based face processing to implement and ensure no raw images are retained.
- System integration – Integrate the age detection module with the real-time advertisement recommendation engine.
- Testing and tuning – Conduct field trials to evaluate user engagement, detection of ages, and relevance of recommendations.
- Evaluation – Compare to baseline (static or passive systems) using quantitative

measures such as detection accuracy, dwell time, and advertising effect.

3.4.Tasks and Sub-Tasks Identified to Meet Objectives

1. Game interface development
 - Design interactive mini-game components.
 - Implement camera capture during playing.
 - Test usability to make users play voluntarily.
2. Age detection model development
 - Select architecture (e.g., CNN or light-weight deep model).
 - Preprocess datasets (face cropping, alignment, augmentation).
 - Train and validate the model with high-quality images.
 - Speed up inference for real-time deployment.
3. Privacy-preserving mechanisms
 - Facial embedding generation implementation.
 - Avoid storage of any raw images within the system pipeline.
 - Validate for GDPR compliance and data retention at minimum.
4. Integration of recommendation engine
 - Create back-end service for real-time ad selection.
 - Integrate age detection output with dynamic ad display logic.
 - Test for perfect operation under different demographic conditions.
5. System testing and evaluation

- Perform controlled laboratory tests to determine accuracy and latency.
- Conduct pilot trials in a retail-like setup to determine engagement and ad performance.
- Collect user feedback to finalize the interface and determine acceptance.

3.5.Data Requirements and Collection Methods

The module requires a high-resolution, heterogeneous facial image dataset for training and testing the age detection model. The dataset should:

- Have broad age range coverage (children, youth, adults, seniors).
- Be demographically diverse for fairness and generalizability.
- Represent real-world conditions (lighting conditions, occlusion, pose, and facial expressions).

Data sources may include publicly available datasets (e.g., UTKFace, IMDB-WIKI) and controlled data from pilot studies. Data will be preprocessed using face detection, alignment, and augmentation (rotation, cropping, brightness adjustment) for increased robustness. All captured test data will be anonymized by using embeddings rather than storing original images.

3.6.Project Requirements

3.6.1. Functional Requirements

- **Identify age accurately from real-time facial inputs.**

The system will need to use advanced computer vision and deep learning algorithms to guess the age of individuals in real time using facial photos from normal cameras. The detection model must be trained on a massive and varied dataset so that the system can withstand changes in lighting, pose, facial expressions, and occlusions. This functionality will be required to make sure that ads delivered are targeted correctly to the relevant age groups, engaging them effectively and achieving relevance.

- **Highly integrated with the recommendation engine.**

The module must provide accurate age predictions as machine-readable information that can be consumed easily by the recommendation engine to select and render contextual ads. The integration must be seamless and run in real time, so the system is able to operate under zero latency in dynamic settings such as shopping malls. Any changes in age distributions perceived should instantaneously result in recommendations generated by the system.

- **Support live monitoring and batch analysis.**

The system should enable support for real-time detection of age in live monitoring and ability to work with recorded video or image data for performance testing and model training. The dual-mode feature ensures that the solution performs adequately in live application while simultaneously allowing for offline validation and calibration.

- **Automatically process multiple faces at once.**

The system must be able to identify and process several faces that appear within one frame to give accurate and just estimation of age groups for crowded scenes. This requirement is crucial to avoid losing possible customers during a scenario with large crowds in a store.

- **Detect age accurately from real-time facial inputs.**

The system must implement advanced computer vision and deep learning algorithms to estimate the age of individuals in real time using facial images captured through standard cameras. The detection model should be trained on a large and diverse dataset to ensure robustness against variations in lighting, pose, facial expressions, and occlusions. This functionality is essential for ensuring that advertisements displayed are appropriately targeted to the correct age groups, thereby maximizing engagement and

relevance.

- **Integrate seamlessly with the recommendation engine.**

The component must provide accurate age predictions as structured data that can be readily used by the recommendation engine to select and display targeted advertisements. This integration should be smooth and operate without latency to ensure real-time performance in dynamic environments such as shopping malls. Any changes in detected age distributions should instantly reflect in the recommendations generated by the system.

- **Support live monitoring and batch analysis.**

The system should support both real-time age detection for active monitoring and the ability to analyze recorded video or image data for performance evaluation and model improvement. This dual-mode capability ensures that the solution is effective in live deployments while also enabling offline validation and tuning.

- **Automatically handle multiple faces simultaneously.**

The system must detect and process multiple faces appearing in a single frame, ensuring fair and accurate age group estimation for crowded scenarios. This requirement is critical to avoid missing potential customers when large groups are present in a retail space.

3.6.2. User Requirements

- **Provide a simple-to-use interface for non-technical users.**

The age detection system must have a simple interface that makes it easy for mall managers and advertisers to monitor detected age demographics without technical know-how. The system must provide clear visualizations such as age group distributions, system performance metrics, and detection statistics in real time.

- **Give rapid results with simple setup.**

The users must be able to deploy and utilize the system with minimal configuration. The hardware must be ordinary, with only plain surveillance cameras and general computing infrastructure, without specialized hardware. Outputs must be given in a timely fashion to offer no lag in advertisement targeting.

- **Ensure transparency and trust.**

The solution should allow users to understand how age detection results are generated without revealing raw personal information. Accuracy levels, detection confidence, and possible limitations should be described so that users will perceive the suggestions provided by the system as credible.

3.6.3. System Requirements

- **Good computational efficiency with inference speed optimized.**

The system must support doing good age estimation with little latency on mid-range hardware. The base model must be performance-optimized through techniques like model quantization, pruning, or hardware acceleration to support real-time capability within the product pipeline.

- **Compatibility with standard video input formats.**

The module should be able to receive inputs from commonly used CCTV or IP camera formats that support various resolutions and frame rates without the use of expensive custom cameras. Integration with existing mall security systems will reduce installation cost and complexity.

- **Large-scale deployable architecture.**

The solution must be scalable across multiple devices or cameras to cover large retail spaces. Both distributed edge processing and centralized monitoring must be achievable, so the solution can be tailored for deployments in a range of sizes and complexity.

3.6.4. Non-Functional Requirements

- **Accuracy and Robustness.**

The model must deliver uniform performance across the different demographic segments, with minimal biases. Accuracy requirements must be higher than the industry norms, and the system must be resilient even in challenging conditions such as low illumination or partially occluded faces.

- **Data Privacy**

The solution must not hold raw user images or personally identifiable information (PII). Anonymized embeddings or only aggregated data should be processed, thus following privacy legislation such as GDPR. This maintains user trust and complies with ethical requirements.

- **Reliability**

The element must operate continuously without repeated shutdown. Failover support, automatic recovery on system crash, and error logging must be integrated to allow for non-stop service in high-volume environments such as shopping centers.

- **Flexibility**

The system architecture should accommodate easy updating and integration of newer models or additional detection capabilities. This allows the component to evolve in parallel with technology developments without requiring complete overhaul.

3.6.5. Use Case Diagram (tentative)

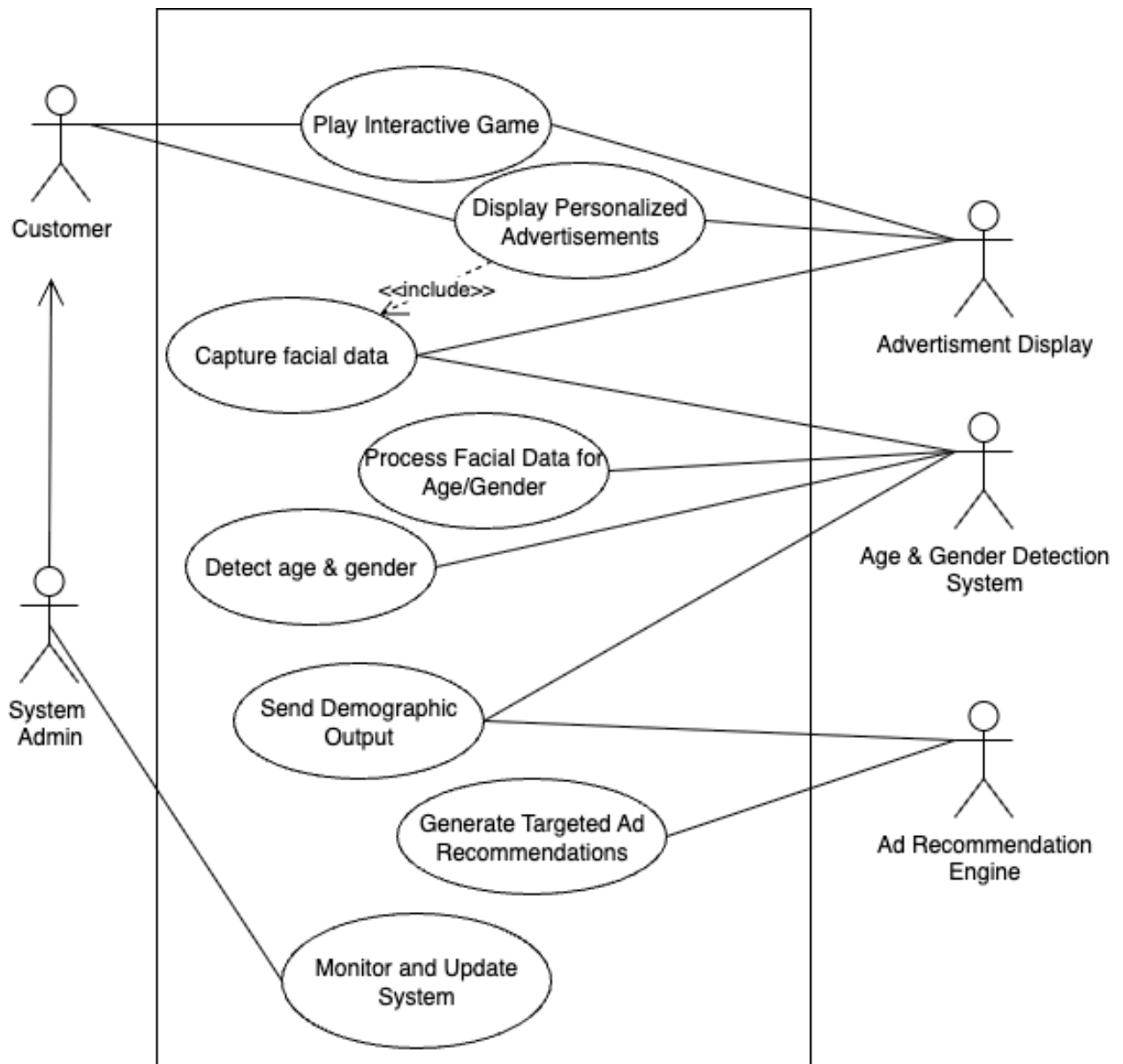


Figure 3-Interactive Game-Based Age and Gender Detection for Targeted Advertising Use Case Diagram

3.6.6. Test Cases (tentative)

Table 3- Test Cases (Tentative)

Test Area	Test Case ID	Description	Input Data	Expected Output
1. Game Interaction	TC-1.1	Verify game starts correctly when user interacts	User taps “Start”	Game interface loads instantly
	TC-1.2	Check responsiveness of game controls	Various user gestures (tap, swipe)	Game reacts within <100ms
	TC-1.3	Validate scoring mechanism	Correct/incorrect moves	Points updated accurately
	TC-1.4	Ensure game ends gracefully	Game timeout or user quit	Proper end screen with summary
2. Face Capture & Quality	TC-2.1	Verify camera activation on game start	User starts game	Camera opens automatically
	TC-2.2	Validate proper lighting detection	Dim / bright environments	System warns if lighting insufficient
	TC-2.3	Test face positioning prompt	User moves off-center	System instructs repositioning
	TC-2.4	Verify frame capture during play	Continuous gameplay	High-quality face frames stored
3. Age & Gender Detection Accuracy	TC-3.1	Validate correct age prediction for young users	Face of age ~10	System predicts child age range
	TC-3.2	Validate correct age prediction for adults	Face of age ~25	System predicts adult age range
	TC-3.3	Verify gender detection accuracy (male)	Male face input	Gender classified as male

	TC-3.4	Verify gender detection accuracy (female)	Female face input	Gender classified as female
4. Data Handling & Privacy	TC-4.1	Verify no raw image storage beyond session	Game played once	Images deleted after processing
	TC-4.2	Validate anonymized embedding storage	Face processed	Only embeddings saved, not raw images
	TC-4.3	Ensure secure transmission to ad module	Detection output sent	Data encrypted in transit
	TC-4.4	Confirm compliance with consent screen	User denies camera access	Game blocks entry gracefully

3.7. Proposed Project Execution

Phase 1: Requirement Analysis & Literature Review

- Determine system goals: gamified real-time gender and age recognition.
- Review existing research on demographic identification, gamified HCI, and ad targeting.
- Define functional, non-functional, and user requirements specific to the component.

Phase 2: Data Collection & Preprocessing

- Obtain face datasets for some age groups and genders.
- Perform preprocessing: face alignment, normalization, resizing, and augmentation for increased robustness.
- Prepare logs of game interaction data for measuring engagement levels.

Phase 3: Model Training & Development

- Select light-weight CNN models (MobileNet, ResNet-18, or EfficientNet) for real-time deployment.
- Train models through transfer learning with face embeddings for age group and gender identification.
- Test models on benchmark **datasets and optimize for accuracy and latency.**

Phase 4: Integration with Game Interface

- Design and develop the light-weight interactive game (touch-based) to encourage voluntary use.
- Incorporate camera capture into the game framework.
- Provide smooth interaction between gameplay, face detection, and demographic analysis modules.

Phase 5: System Integration & Testing

- Integrate the component with the ad recommendation engine for real-time delivery of target ads.
- Test in real-world conditions (different lighting, postures, and crowd densities).
- Conduct evaluation on accuracy, response time, and user engagement.
- Employ privacy-preserving methods: store only demographic embeddings, not original images.

Phase 6: Evaluation & Refinement

- Measure success against expected KPIs:

Table 4 - Measurement KPIs

KPI	Value
Age Detection Accuracy	$\geq 75\%$
Gender Detection Accuracy	$\geq 85\%$
Latency	≤ 1 second per frame
User Engagement	$\geq 70\%$ participation rate during pilot testing

- Refine game mechanics and model performance based on feedback.
- Document findings for final system integration into the overall research project

3.8.Mathematical Formulation

The first step of the system is demographic feature extraction (age and gender) from facial inputs acquired using the interactive game-based interface. This module employs deep learning models to learn stable embeddings and classify them.

Feature Extraction (Face Embedding):

Given an input face image x , a convolutional neural network (CNN) such as ResNet or MobileNet is used to embed the face in a low-dimensional feature vector:

$$e = f_{\theta}(x)$$

where e represents the embedding, and f_{θ} is the CNN parameterized by θ . This embedding captures discriminative facial characteristics.

Gender Classification

Gender is treated as a binary class problem. A logistic regression layer is employed on top of the embedding to predict the probability of the input being in the male class:

$$\hat{p}_{\text{male}} = \sigma(g\theta_g(e)); \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

The associated binary cross-entropy loss is:

$$\mathcal{L}_{\text{gender}} = -[y_{\text{gender}} \log(\hat{p}_{\text{male}}) + (1 - y_{\text{gender}}) \log(1 - \hat{p}_{\text{male}})]$$

This assists the model to detect male and female users correctly in real-time.

Age Estimation:

There are two approaches: regression and classification. For public mall environment robustness, we use age-group classification, where pre-defined age bins (e.g., child, teenager, adult, senior) are utilized. The softmax probability for each class is:

$$\hat{p}_c = \frac{\exp(h_c(e))}{\sum_{k \in \mathcal{C}} [\exp(h_k(e))]}$$

with the categorical cross-entropy loss:

$$\mathcal{L}_{\text{age}} = - \sum_{c \in \mathcal{C}} y_c \log(\hat{p}_c)$$

This method makes the system more reliable in case precise age cannot be forecast.

Joint Multi-Task Learning:

Since both age and gender share the same embeddings, a multi-task approach is adopted.

The joint optimization task is:

$$\mathcal{L}_{AG} = \lambda_{\text{age}} \mathcal{L}_{\text{age}} + \lambda_{\text{gender}} \mathcal{L}_{\text{gender}}$$

where λ_{age} and λ_{gender} balance the relative importance of each task. This allows the system to efficiently learn complementary features for demographic detection while reducing redundancy.

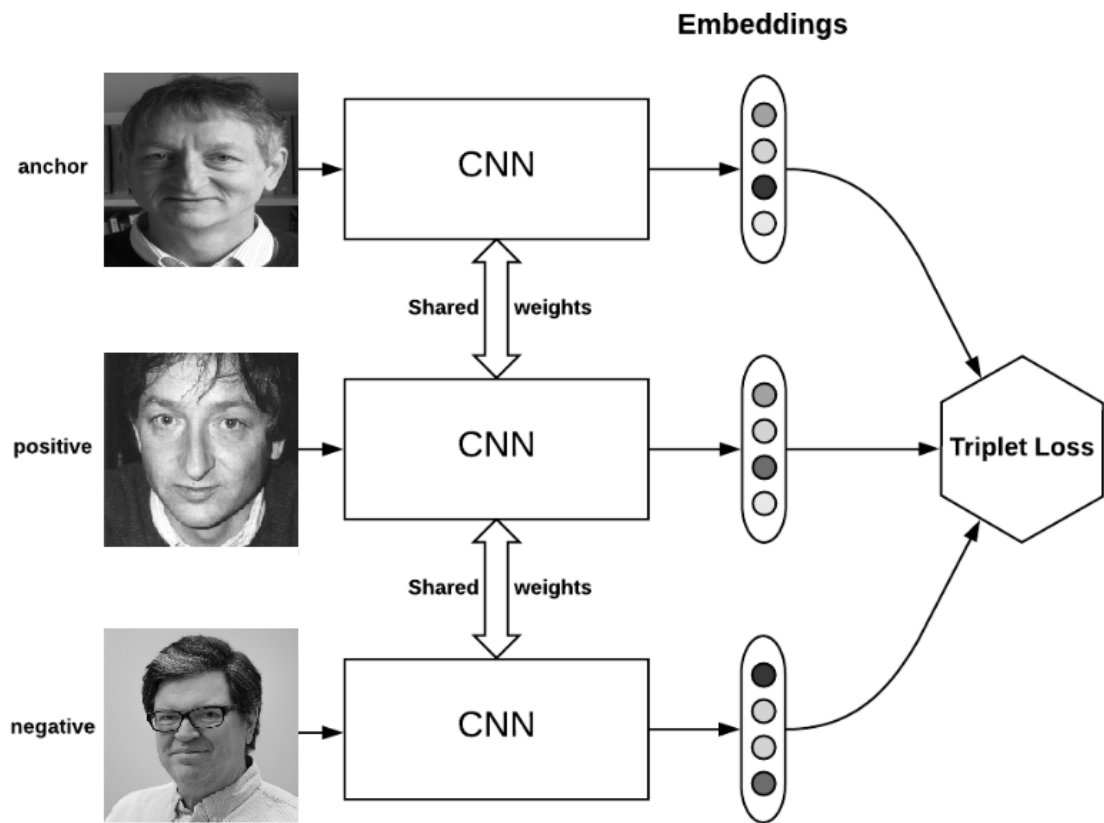


Figure 4-Illustration of Triplet Loss function showing Anchor, Positive, and Negative embeddings. Source: Adapted from [22].

3.9. Anticipated Conclusion

By following this strategy, the project will deliver an interactive, privacy-preserving advertisement module capable of delivering dynamic, age-targeted content in real time. The incorporation of gamification and deep learning will improve both user engagement and detection accuracy compared to traditional static systems. The system should demonstrate:

- Higher-quality facial data due to active user participation.
- Enhanced personalization of ads using demographic information.
- Adherence to privacy legislation through data processing without storing raw facial images.

This research will validate how interactive advertising can balance effectiveness, user experience, and responsible data handling, offering a sustainable solution for today's retail environments

4. DESCRIPTION OF PERSONAL & FACILITIES

Supervisor – Dr. Bhagya Nathali Silva, Assistant Professor, Faculty of Computing, Sri Lanka Information Technology.

Co – Supervisor – Ms. Shashini Kumarasinghe, Assistant Lecturer, Faculty of Computing, Sri Lanka Information Technology.

The project contributions are a holistic approach to real-time advertising and security enhancement through combined age, emotion, and behavior recognition systems.

Fonseka W. S. M. focuses on building the interactive game-based age recognition subsystem, dividing users accurately into age categories to help deliver targeted advertising. Samaraweera W D U I oversees the emotion detection module, detecting risky emotional indicators such as fear, stress, or aggression and keeping a list of potentially risky individuals. Abeykoon S N oversees the behavior detection capability, monitoring individuals' actions to validate the suspicions raised and sending safety alerts upon verification of high-risk behavior. Rajana H T R trains and deploys the recommendation engine based on join output from age, emotion, and behavior inference to deliver accurate, privacy-protecting advertising to various audiences in real time.

4.1. Facilities

The project utilizes the computational resources, including high-performance workstations with GPU capability for model testing and training. Commercially available surveillance cameras and edge light devices simulate a normal real-time shopping mall environment. Software tools utilized are Python-based machine learning libraries (TensorFlow, PyTorch), real-time video processing toolkits (OpenCV, MediaPipe), and cloud platforms for model serving and testing.

This collaborative setup ensures seamless integration of all components, providing a robust and scalable system that can enhance both personalized advertising and mall security management simultaneously.

5. BUDGET & BUDGET JUSTIFICATION

The estimated budget for the project primarily covers the hardware costs need and the subscription costs. We do not aim to commercialize our individual components but the whole research will be commercialized as a single project. Following is the budget justification for the whole research system.

Purpose	Cost (LKR)
Hardware Cost	30000
Software & Subscription Cost	6000
Travelling Cost	5000
Total	41000

6. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed *Interactive Game-Based Age Detection for Targeted Advertising* system is designed to cater to individuals across all age categories, providing advertisers with a dynamic and highly precise tool for delivering personalized promotions in real time. By combining engaging human–computer interaction with advanced computer vision, this solution stands out as both a user-friendly and business-oriented innovation. Its commercial value lies in enabling advertisers to maximize engagement, reduce ad fatigue, and increase conversion rates by showing contextually relevant advertisements to mall visitors without collecting personally identifiable information. The product is adaptable to multiple retail and entertainment environments, ranging from shopping malls and department stores to event venues and airports, offering a broad and sustainable commercialization potential.

5.1. Market Space

1. Advertising Market

Targeted advertising and digital signage are growth markets, with global digital-out-of-home (DOOH) advertising set to grow at double-digit percentages per year. With businesses pursuing measurable and addressable engagement, there is growing demand for AI-driven systems that show ads responsibly based on audience characteristics. Systems that work interactively and engage users actively rather than being passively available tend to align with market trends, and hence this solution is timely and apt.

2. Market Size

The market for targeted advertising and retail analytics is worth a couple of billion dollars globally, with strong growth rates in the Asia-Pacific region where shopping malls remain the prime retail channel. While in Sri Lanka and other developing economies the widespread uptake of AI-based targeting systems remains in its embryonic stages, there is considerable enthusiasm among advertisers to invest in solutions that boost campaign efficacy without violating privacy regulations. This offers

a suitable entry point for low-cost, cutting-edge, and adaptive technologies such as this system.

3. Existing Systems

Current systems rely primarily on static camera-based facial recognition and demographic estimation models integrated in electronic displays. While the existing systems provide minimal targeting capabilities, they are riddled with problems of low accuracy in uncontrolled environments, inability to actively interact with users, and greater public concern regarding privacy. Current solutions are also expensive and not scalable for small and medium-sized enterprises, creating room for low-cost yet user-friendly solutions.

4. Competition

The market is dominated by established DOOH tech companies and companies that offer state-of-the-art demographic detection solutions. There are, however, very few that combine real-time facial analysis with a gamification component to actively collect high-quality data and enrich user experience. The product is differentiated on the grounds of interactivity, privacy-centric design (using embeddings rather than storing raw images), and a scalable architecture that can function optimally in edge devices. These qualities deliver a competitive advantage through cost savings in infrastructure as well as improved detection accuracy and customer acceptance.

5.2. Revenue Model and Commercial Strategy

1. **Licensing Model** – Retail chain, ad agencies, or owners of shopping malls can purchase licenses to deploy the system across numerous locations. This model provides stable and predictable revenue and is readily scalable with client expansion.
2. **Subscription Model (Software-as-a-Service)** – Offering the platform as a subscription model will make the solution within the budget of even small retailers with low upfront cost. Regular updates, analytics dashboard, and support

services can be offered as premium plans.

3. Advertising Partnerships – Collaborating with brands to deliver targeted advertisements directly via our system presents another revenue source. Advertisers can be charged on a per impression basis, per engagement, or based on dwell time metrics recorded by the system.
4. Hardware + Software Bundling – For locations that lack an installed infrastructure, package deals that pair low-cost display units or kiosks with the software can open new markets while providing a turnkey solution.
5. Data-Driven Insights as a Service – Aggregated demographic and age information, without retaining personal details, can be sold as anonymized analytics to retailers and marketing companies to optimize campaign strategies.

This multi-pronged commercialization strategy ensures that the product can be offered to a broad range of customers—from small retailers to large shopping mall chains—while being low-cost, scalable, and having strong privacy safeguards.

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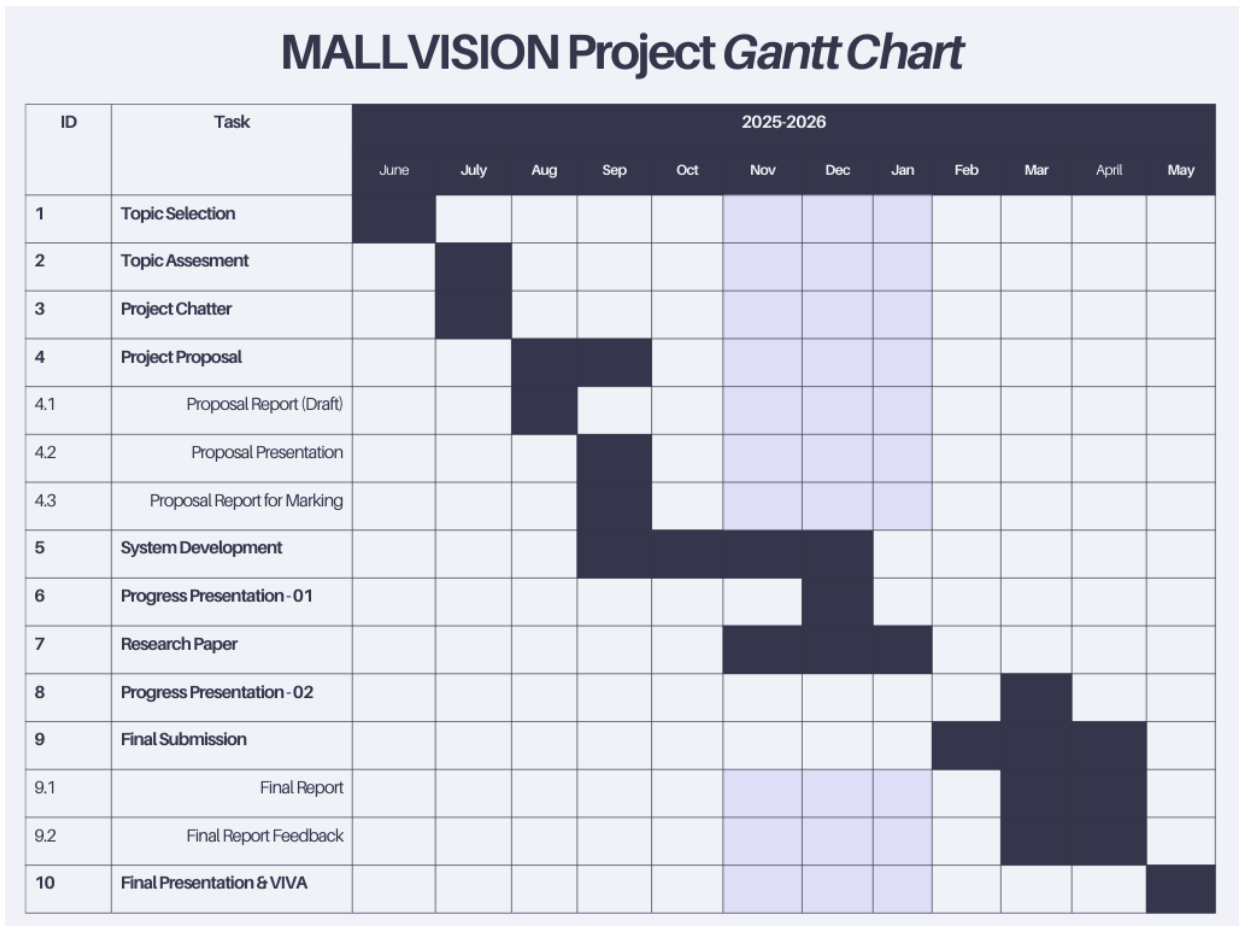
APPENDICES

APPENDIX 1 – Work Breakdown Structure



Appendix 1 - Work Breakdown Structure

APPENDIX 2 – Gantt Chart



Appendix 2 - Gantt Chart